

Solution for the Intersection of Two Hyperbolas

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Question:

Solve the following system of rational equations

$$\frac{10}{x+y} + \frac{2}{x-y} = 4 \quad (0.1)$$

$$\frac{15}{x+y} - \frac{5}{x-y} = -2 \quad (0.2)$$

Solution:

We convert the system of rational equations from (0.1) and (0.2) into the standard quadratic form for conic sections. For the first equation (0.1), we multiply by $(x+y)(x-y)$:

$$\begin{aligned} 10(x-y) + 2(x+y) &= 4(x^2 - y^2) \\ 12x - 8y &= 4x^2 - 4y^2 \\ \implies x^2 - y^2 - 3x + 2y &= 0 \end{aligned} \quad (0.3)$$

For the second equation (0.2), we do the same:

$$\begin{aligned} 15(x-y) - 5(x+y) &= -2(x^2 - y^2) \\ 10x - 20y &= -2x^2 + 2y^2 \\ \implies x^2 - y^2 + 5x - 10y &= 0 \end{aligned} \quad (0.4)$$

The equations for the hyperbolas can be written in the form $\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0$, where $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$. For hyperbola H_1 from (0.3):

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 2 \begin{pmatrix} -3/2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 0 = 0 \quad (0.5)$$

For hyperbola H_2 from (0.4):

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 2 \begin{pmatrix} 5/2 & -5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 0 = 0 \quad (0.6)$$

In homogeneous coordinates, using the form $\mathbf{x}^T \mathbf{C} \mathbf{x} = 0$, where $\mathbf{x} = \begin{pmatrix} x & y & 1 \end{pmatrix}^T$, the

matrices for the hyperbolas are:

$$C_1 = \begin{pmatrix} 1 & 0 & -3/2 \\ 0 & -1 & 1 \\ -3/2 & 1 & 0 \end{pmatrix} \quad (0.7)$$

$$C_2 = \begin{pmatrix} 1 & 0 & 5/2 \\ 0 & -1 & -5 \\ 5/2 & -5 & 0 \end{pmatrix} \quad (0.8)$$

The intersection points lie on a degenerate conic formed from a linear combination $H_1 + \mu H_2 = 0$. We must find the value of μ that makes the determinant of the combined conic's matrix, $C(\mu) = C_1 + \mu C_2$, equal to zero. The general matrix for this combination is:

$$C(\mu) = \begin{pmatrix} 1 + \mu & 0 & \frac{5\mu-3}{2} \\ 0 & -(1 + \mu) & 1 - 5\mu \\ \frac{5\mu-3}{2} & 1 - 5\mu & 0 \end{pmatrix} \quad (0.9)$$

We find the determinant by reducing the matrix to row echelon form.

$$C(\mu) \xrightarrow{R_3 \rightarrow R_3 - \frac{5\mu-3}{2(1+\mu)} R_1} \begin{pmatrix} 1 + \mu & 0 & \frac{5\mu-3}{2} \\ 0 & -(1 + \mu) & 1 - 5\mu \\ 0 & 1 - 5\mu & -\frac{(5\mu-3)^2}{4(1+\mu)} \end{pmatrix} \quad (0.10)$$

$$\xrightarrow{R_3 \rightarrow R_3 + \frac{1-5\mu}{1+\mu} R_2} \begin{pmatrix} 1 + \mu & 0 & \frac{5\mu-3}{2} \\ 0 & -(1 + \mu) & 1 - 5\mu \\ 0 & 0 & \frac{5(15\mu^2-2\mu-1)}{4(1+\mu)} \end{pmatrix} \quad (0.11)$$

The determinant is the product of the diagonal entries. We set it to zero:

$$\begin{aligned} \det(C(\mu)) &= (1 + \mu) \cdot (-(1 + \mu)) \cdot \left(\frac{5(15\mu^2 - 2\mu - 1)}{4(1 + \mu)} \right) = 0 \\ \implies & -\frac{5(1 + \mu)(15\mu^2 - 2\mu - 1)}{4} = 0 \end{aligned} \quad (0.12)$$

This equation is satisfied if either $(1 + \mu) = 0$ or $(15\mu^2 - 2\mu - 1) = 0$. The simplest root is obtained from the first factor, which gives $\mu = -1$. The equation for this degenerate conic is found by subtracting (0.4) from (0.3):

$$\begin{aligned} (x^2 - y^2 - 3x + 2y) - (x^2 - y^2 + 5x - 10y) &= 0 \\ -8x + 12y &= 0 \\ \implies y &= \frac{2}{3}x \end{aligned} \quad (0.13)$$

We find the intersection of the common chord from (0.13) with H_1 using the formula $\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m}$. The line and conic parameters from H_1 are: $\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\mathbf{m} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$, $\mathbf{V} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\mathbf{u} = \begin{pmatrix} -3/2 \\ 1 \end{pmatrix}$, and $f = 0$.

Next, we compute the components for the parameter κ_i :

$$\mathbf{m}^\top \mathbf{V} \mathbf{m} = 5 \quad (0.14)$$

$$g(\mathbf{h}) = 0 \quad (0.15)$$

$$\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) = -\frac{5}{2} \quad (0.16)$$

We substitute these into the formula for κ_i :

$$\begin{aligned} \kappa_i &= \frac{1}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \left(-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h})(\mathbf{m}^\top \mathbf{V} \mathbf{m})} \right) \\ &= \frac{1}{5} \left(-\left(-\frac{5}{2}\right) \pm \sqrt{\left(-\frac{5}{2}\right)^2 - (0)(5)} \right) = \frac{1}{5} \left(\frac{5}{2} \pm \frac{5}{2} \right) \end{aligned} \quad (0.17)$$

This gives two values: $\kappa_1 = 1$ and $\kappa_2 = 0$.

Finally, we find the intersection points using $\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m}$:

$$\mathbf{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \quad (0.18)$$

$$\mathbf{x}_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (0.19)$$

The intersection points are $(3, 2)$ from (0.18) and $(0, 0)$ from (0.19). The point $(0, 0)$ is extraneous because it causes division by zero in the original equations (0.1) and (0.2). Thus, the only valid solution is $x = 3, y = 2$.

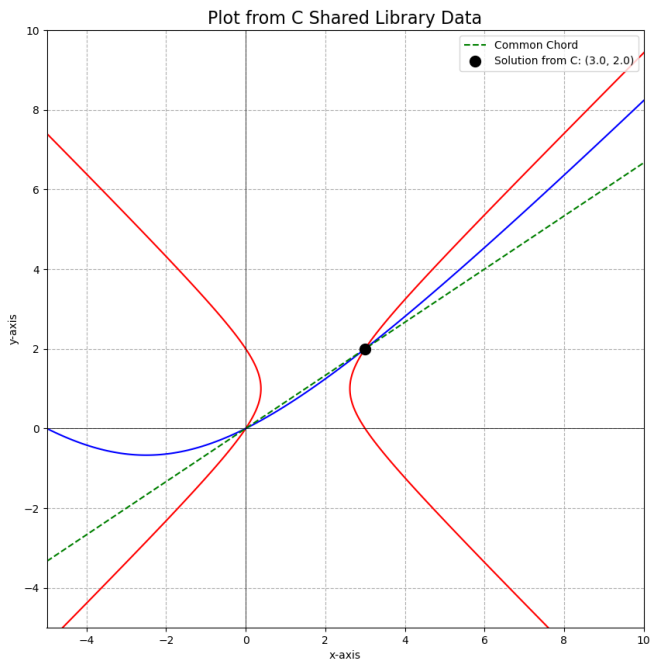


Fig : Hyperbolas

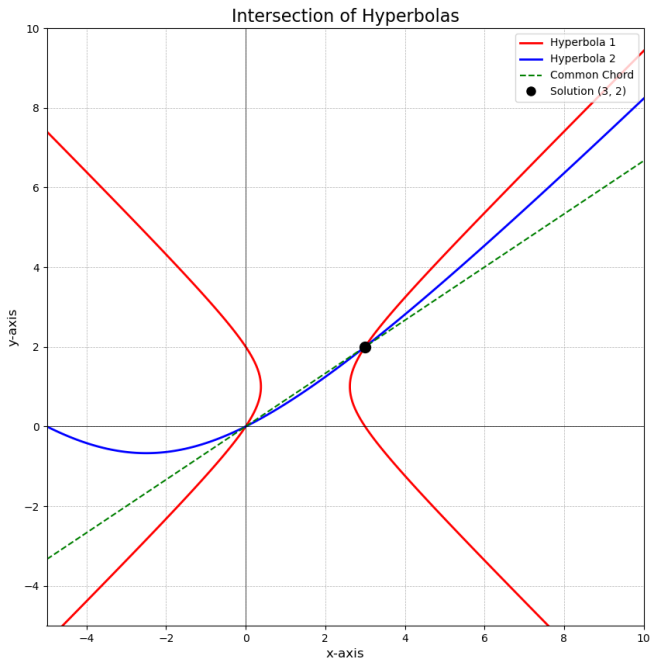


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